Assignment 5 Solutions

5.22

$$\begin{bmatrix} 1 & 1 & 2 \\ 2^{-2} & 3^{-2} & 5^{-4} \\ {-1}^{+1} & {-3}^{+1} & {-3}^{+2} \end{bmatrix} \sim \begin{bmatrix} 1 & 1 & 2 \\ 0 & 1 & 1 \\ 0 & {-2}^{+2} & {-1}^{+2} \end{bmatrix} \sim \begin{bmatrix} \boxed{1} & 1 & 2 \\ 0 & \boxed{1} & 1 \\ 0 & 0 & \boxed{1} \end{bmatrix}$$

Both 1-1 and onto. 1 pivot per. col and row

5.23

$$\begin{bmatrix} 5 & 0 & 5 \\ 0 & 0 & 0 \\ 4 & 0 & 7 \end{bmatrix}$$

Neither. Columns are not L.I., at most 2 pivots per row.

5.25

$$\begin{bmatrix} \boxed{1} \\ 1 \\ 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

1-1, not onto. 1 pivot and 1 column but 6 rows

5.27

$$\begin{bmatrix} 1 & 2 & -1 & 6 & 9 \\ 2^{-2} & 5^{-4} & {-5}^{+2} & 13^{-12} & 22^{-18} \\ {-3}^{+3} & {-4}^{+6} & {-3}^{-3} & {-16}^{+18} & {-19}^{+27} \end{bmatrix} \sim \begin{bmatrix} 1 & 2 & -1 & 6 & 9 \\ 0 & 1 & -3 & 1 & 4 \\ 0 & 2 & -6 & 2 & 8 \end{bmatrix} \sim \begin{bmatrix} \boxed{1} & 2 & -1 & 6 & 9 \\ 0 & \boxed{1} & -3 & 1 & 4 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Neither. 2 pivots, 3 rows and 5 columns

6.2

$$2\begin{bmatrix} 6 & 1 \\ -1 & -3 \\ 8 & 1 \end{bmatrix} - 4\begin{bmatrix} 6 & -5 \\ -6 & -2 \\ -3 & 4 \end{bmatrix} - 3\begin{bmatrix} -4 & 0 \\ -8 & 5 \\ -4 & -1 \end{bmatrix} =$$

$$\begin{bmatrix} 12 & 2 \\ -2 & -6 \\ 16 & 2 \end{bmatrix} - \begin{bmatrix} 24 & -20 \\ -24 & -8 \\ -12 & 16 \end{bmatrix} - \begin{bmatrix} -12 & 0 \\ -24 & 15 \\ -12 & -3 \end{bmatrix} =$$

$$\begin{bmatrix} (12-24+12) & (2+20) \\ (-2+24+24) & (-6+8-15) \\ (16+12+12) & (2-16+3) \end{bmatrix} = \boxed{\begin{bmatrix} 0 & 22 \\ 46 & -13 \\ 40 & -11 \end{bmatrix}}$$

6.5

$$\begin{bmatrix} -2 & 6 \\ -2 & -7 \\ -3 & -2 \end{bmatrix}\begin{bmatrix} -4 & 8 & 4 \\ -10 & -4 & 9 \end{bmatrix} = \begin{bmatrix} (-2(-4)+6(-10)) & (-2(8)+6(-4)) & (-2(4)+6(9)) \\ (-2(-4)-7(-10)) & (-2(8)-7(-4)) & (-2(4)-7(9)) \\ (-3(-4)+(-2)(-10)) & (-3(8)+(-2)(-4)) & (-3(4)+(-2)(9)) \end{bmatrix}$$

$$= \begin{bmatrix} -52 & -40 & 46 \\ 78 & 12 & -71 \\ 32 & -16 & -30 \end{bmatrix}$$

6.6

$$\begin{bmatrix} 8 & 6 & -1 & -2 \end{bmatrix}\begin{bmatrix} 3 \\ 5 \\ 1 \\ 5 \end{bmatrix} = 8(3)+6(5)-1(1)-2(5) =$$

$$24 + 30 - 1 - 10 =$$

$$\boxed{43}$$

6.7

$$\begin{bmatrix} -3 \\ 6 \\ 8 \\ 5 \end{bmatrix}\begin{bmatrix} -9 & -7 & 0 & -8 \end{bmatrix} = \boxed{\begin{bmatrix} 27 & 21 & 0 & 24 \\ -54 & -42 & 0 & -48 \\ -72 & -56 & 0 & -64 \\ -45 & -35 & 0 & -40 \end{bmatrix}}$$

6.8 $\begin{bmatrix} 7 & -10 \\ -6 & 6 \\ 1 & 0 \\ -4 & 2 \end{bmatrix} \begin{bmatrix} -8 & 9 \\ -10 & 0 \\ -9 & 1 \\ -4 & -3 \end{bmatrix}$ Not well-defined

$\mathbb{R}^{4\times\boxed{2}}$ $\quad$ $\mathbb{R}^{\boxed{4}\times 2}$

do not match

6.18 $$\begin{bmatrix} \cos 2 & -\sin 2 & 0 \\ \sin 2 & \cos 2 & 0 \\ 0 & 0 & 1 \end{bmatrix}^{2023} = \begin{bmatrix} \cos 4046 & -\sin 4046 & 0 \\ \sin 4046 & \cos 4046 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

This is equivalent to 2023 rotations around the z-axis by 2 rad.

5.37 True. It's the only way to has 1 pivot per column <u>and</u> row

5.38 False. $\begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$

5.39 False. $\vec{x} \mapsto \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix} \vec{x}$

6.33 True. Stated in lecture, matrix addition is elementwise.

6.36 False. $A = B = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

6.42 **False.**

$$\begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 2 & 2 \end{bmatrix}$$

$$\neq$$

$$\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}\begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 1 & 2 \end{bmatrix}$$

6.47 **False.**

$$\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}\begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

6.49 **False.**

$$\begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}\begin{bmatrix} 1 & 0 \\ 0 & \frac{1}{2} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & \frac{1}{2} \end{bmatrix}\begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

5.57 a)

$$S(\vec{x}) = \begin{bmatrix} 1 & 0 \\ 0 & 2 \\ 1 & -1 \end{bmatrix}\vec{x} \qquad T(\vec{x}) = \begin{bmatrix} 1 & 1 & 1 \\ -1 & -2 & -3 \end{bmatrix}$$

$$S \circ T(\vec{x}) = \begin{bmatrix} 1 & 0 \\ 0 & 2 \\ 1 & -1 \end{bmatrix}\begin{bmatrix} 1 & 1 & 1 \\ -1 & -2 & -3 \end{bmatrix}\vec{x}$$

$$= \boxed{\begin{bmatrix} 1 & 1 & 1 \\ -2 & -4 & -6 \\ 2 & 3 & 4 \end{bmatrix}}\vec{x}$$

b)

$$\begin{bmatrix} 1 & 1 & 1 & 1 \\ \overset{+2}{-2} & \overset{+2}{-4} & \overset{+2}{-6} & \overset{+2}{8} \\ \overset{-2}{2} & \overset{-2}{3} & \overset{-2}{4} & \overset{-2}{-3} \end{bmatrix} \sim \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & -2 & -4 & 10 \\ 0 & 1 & 2 & -5 \end{bmatrix} \sim \begin{array}{c} \begin{array}{cccc} & -1 & -2 & +5 \end{array} \\ \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 2 & -5 \\ 0 & 0 & 0 & 0 \end{bmatrix} \end{array}$$

$$\sim \begin{bmatrix} 1 & 0 & -1 & 6 \\ 0 & 1 & 2 & -5 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$x_1 = 6 + x_3$

$x_2 = -5 - 2x_3$

x_3 is free

6.70

$$\begin{bmatrix} 1 & -1 & 2 & 1 \\ \overset{+3}{-3} & \overset{-3}{4} & \overset{+6}{2} & \overset{+3}{0} \end{bmatrix} \sim \begin{bmatrix} 1 & \overset{+1}{-1} & \overset{+8}{2} & \overset{+3}{1} \\ 0 & 1 & 8 & 3 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 10 & 4 \\ 0 & 1 & 8 & 3 \end{bmatrix}$$

$$\vec{b}_1 = \begin{bmatrix} 4 \\ 3 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & -1 & 2 & 0 \\ \overset{+3}{-3} & \overset{-3}{4} & \overset{+6}{2} & 1 \end{bmatrix} \sim \begin{bmatrix} 1 & \overset{+1}{-1} & \overset{+8}{2} & \overset{+1}{0} \\ 0 & 1 & 8 & 1 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 10 & 1 \\ 0 & 1 & 8 & 1 \end{bmatrix}$$

$$b_2 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \qquad B = \begin{bmatrix} 4 & 1 \\ 3 & 1 \\ 0 & 0 \end{bmatrix}$$

* there are multiple possible answers

6.72

$$B \begin{bmatrix} 1 & -1 & 2 \\ -3 & 4 & 2 \end{bmatrix} = \left[B\begin{bmatrix} 1 \\ -3 \end{bmatrix} \quad B\begin{bmatrix} -1 \\ 4 \end{bmatrix} \quad B\begin{bmatrix} 2 \\ 2 \end{bmatrix} \right]$$

$$B = [\vec{b}_1 \;\; \vec{b}_2] \qquad = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \in \text{span}\{\vec{b}_1, \vec{b}_2\} \qquad \Rightarrow \text{span}\{\vec{b}_1, \vec{b}_2\} = \mathbb{R}^3$$

This is not possible.

$$\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \in \text{span}\{\vec{b}_1, \vec{b}_2\}$$

$$\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \in \text{span}\{\vec{b}_1, \vec{b}_2\}$$

6.72 a) $\begin{bmatrix} 1 & -1 & 1 \end{bmatrix} \begin{bmatrix} 1 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -2 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & -1 & 1 \end{bmatrix} \begin{bmatrix} 3 \\ -4 \\ 4 \end{bmatrix}$

$= 3 + 4 + 4 = 11$

b) $\begin{bmatrix} 1 & -1 & 1 & 1 \\ -1 & 2 & -1 & 0 \\ 1 & -2 & 1 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & -1 & 1 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & -1 & 0 & -1 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 1 & 2 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$

$x_1 = 2 - x_3$

$x_2 = 1$

x_3 is free

c) $\vec{v} = \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix}$

$\begin{bmatrix} 0 & 1 & 2 \end{bmatrix} A \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix} =$

$\begin{bmatrix} 0 & 1 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = 0$